



**The Shell Petroleum Development Company of Nigeria
Limited (SPDC)**

Ultimate Recovery Change Report (URCR)

Egbema Field

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1. EXECUTIVE SUMMARY

Egbema field is situated in SPDC's Eastern Division, about 100km NNW of Port Harcourt (Figure 2.1). The field was discovered in October 1960, and started production in April 1970. A peak production of 18,000 bopd was achieved in 1972 and afterwards production declined steadily to 5500 bopd in 1982 when it was shut in due to offtake restrictions. The field was reopened in 1984 and has since been in decline with current (September 2016) production of 783 bopd at 64% water cut and GOR of 1,100 scf/bbl and cumulative production of 77 MMstb of oil and 56 Bscf of gas.

In 2010, the operatorship of the field (along with other fields in Egbema Node) was transferred to the Nigerian Petroleum Development Company (NPDC) based on a side letter to the SPDC JV joint operating agreement (JOA).

In 2016, the operator (NPDC) carried out a field review of opportunities in the field and identified opportunities in the D8000X, D9000X/D9200X and E1000X. This field review then formed the basis of the SPDC Egbema FOD project in which the target reservoirs (D8000X, D9000X, D9100X, D9200X, and E1000X) were seismically re-interpreted and modelled (3D Static model). The D9000, D9100 & D9200 were reviewed and merged into a single reservoir complex (D9000) based on the integration of historical pressure data, contacts and geologic correlation. Hence the D8000X, D9000 Complex and the E1000 are the focus of this URRCR.

Hydrocarbon initially in place volumes re-evaluations for D8000X, D9000 complex and E1000X were estimated in order to ascertain potential incremental volumes per reservoir given cumulative production to date.

Given the conditions above, changes to Stock Tank Oil Initially in Place (STOIIP), Solution Gas Initially in Place (SGIIP), associated gas volumes and Ultimate Recovery volumes are proposed for D8000X, D9000 Complex and E10000 reservoirs.

1.1. Summary of Changes since last ARPR

The key reasons for change in the hydrocarbon initially in place and ultimate recovery for the affected reservoirs from 1/1/2016 to 1/1/2017 include the following:

- Re-processing and re-interpretation of seismic data
- Static geological modeling and volume re-estimation
- Re-evaluation of petrophysical properties
- Re-evaluation of PVT parameters
- Review of technical limit recovery factors based on observed production performance

Table 1 shows the summary of the change and Table 2 list the overview of the reasons for change.

Table 1.1: Summary of changes to P1 and P1+P2 Ultimate Recovery.

	Oil (MMstb)					Solution Gas (Bscf)				
	STOIIP		UR		RF (%)	SGIIP		UR		RF (%)
	P1	P1+P2	P1	P1+P2	P1+P2	P1	P1+P2	P1	P1+P2	P1+P2
01/01/2016	125.0	141.3	41.7	51.1	36	56.8	64.3	43.3	60.3	94
01/01/2017	131.5	182.6	73.5	112.3	62	40.0	80.6	23.2	50.7	63
Change	6.5	41.3	29.5	63.2		-16.8	16.3	-20.1	-9.6	
Change %	5.2	29.2	70.7	123.7		-29.6	25.35	46.42	-15.92	

	Cap Gas (Bscf)					Associated Condensate (MMstb)				
	GIIP		UR		RF (%)	CIIP		UR		RF (%)
	P1	P1+P2	P1	P1+P2	P1+P2	P1	P1+P2	P1	P1+P2	P1+P2
01/01/2016	47.1	57.3	29.8	38.3	67	0.0	0.0	0.0	0.0	NA
01/01/2017	43.7	61.1	14.2	26.4	43	0.3	0.5	0.1	0.2	42
Change	-3.4	4.3	-15.6	-11.9		0.3	0.5	0.1	0.2	
Change %	-7.2	7.5	-52.3	-31.1		-	-	-	-	

	Non-Associated Gas (Bscf)					Non-Associated Condensate (MMstb)				
	GIIP		UR		RF (%)	CIIP		UR		RF (%)
	P1	P1+P2	P1	P1+P2	P1+P2	P1	P1+P2	P1	P1+P2	P1+P2
01/01/2016	0.0	0.0	0.0	0.0	N/A	0.0	0.0	0.0	0.0	N/A
01/01/2017	0.0	0.0	0.0	0.0	N/A	0.0	0.0	0.0	0.0	N/A
Change	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Change %	N/A	N/A	N/A	N/A		N/A	N/A	N/A	N/A	

1.2. Reasons for Change

Table 1.2: Overview of Reasons for Change

		Previous Situation	Current situation	Impact on UR	Impacted Reservoirs
Reason for Change					
New Discoveries		Not applicable	Not applicable	Not applicable	Not applicable
Extensions		Not applicable	Not applicable	Not applicable	Not applicable
Revisions		Not applicable	1994 3D seismic acquisition	Not applicable	Not applicable
	<i>Seismic acquisition</i>				
	<i>Seismic processing and interpretation</i>		2013 seismic reprocessing	Increase	D8000 D9000 E1000
	<i>Petrophysical data</i>		Petrophysical re-evaluation	Increase	D8000 D9000 E1000
	<i>Performance</i>		Improved Performance	Re-evaluation carried out	D8000X, D9000 Complex & E1000X
	<i>Studies</i>		Egbema FOD Pre-DG1 study	Increase	D8000X, D9000 Complex & E1000X
	<i>Fluid data acquisition</i>		None	None	D8000 D9000 E1000
	<i>Well test results</i>		Not applicable	Not applicable	Not applicable
	<i>Methodology changes</i>		None	Not applicable	Not applicable
	<i>Reservoir renaming</i>		D9000 Complex	Increase	
	<i>Other (Specify)</i>		None	None	None
Improved Recovery		Not applicable	Not applicable	Not applicable	Not applicable
Divestments/Acquisitions		Not applicable	Not applicable	Not applicable	Not applicable

3D seismic was acquired across OML 20 in 1994 and processed to Post-Stack Time Migration in 1995 to 1996 by SPDC SIPMAP and Pre-stack Time Migration in 2009. Further re-processing of the 3D seismic was carried out in 2013 to PSDM (Post –Stack Depth Migration) by IDSL for the current studies. This reprocessed Post –Stack Depth Migration (PSDM) seismic data was transferred to SPDC for a re-interpretation of the project target Reservoirs (D8000X, D9000X, complex and E1000X) to validate the interpretation carried out by IDSL on behalf of NPDC .The horizon interpretation by SPDC is quite comparable to that of NPDC for the reservoirs, except in the E1000X reservoir where there was some depth difference in the loop interpreted. SPDC did not carry out a full field Seismic interpretation. The SPDC interpretation of the candidate reservoirs provided the basis for the current field evaluation and development premises.

Depth uncertainties were evaluated for the three reservoirs ranges from (+/-52 ft) for D8000X, D9000X (+/- 41 ft) and E1000X (+/- 78 ft).

1.3. Summary of Principal Uncertainties and Risks to Ultimate Recovery

Table 1.3: Summary of Principal Uncertainties and Risks to Ultimate Recovery

Project	Issue Type	Uncertainties	Risk / Exposure	Mitigation
Egbema	Technical	Structure	Uncertainty in initially in place and recoverable volumes	Low, base and high structural interpretations
	Technical	Properties	Uncertainty in initially in place and recoverable volumes	Low, base and high properties estimated
	Technical	Contacts	Uncertainty in initially in place and recoverable volumes	Low, base and high case contacts estimated as follows: OWC: D8000 • Low/Base/High = Logged D9000 • Low/Base/High = Logged E1000 • Base Case = Estimated based on production data GOC: D8000 • Low /Base/High = Side wall Sample fluorescence D9000 • Low/Base/High = Side wall Sample fluorescence E1000 • Low/Base/high = Side wall Sample fluorescence

2. INTRODUCTION

2.1. Field Overview

Egbema field is situated in SPDC's Eastern Division, about 100km NNW of Port Harcourt (Figure 2.1) and was discovered October 1960. The field comprises 12 identified hydrocarbon reservoirs. It has a total STOIP of about 226 MMSTB and a FGIIIP of about 302 BSCF. The developed reservoirs in the field are D1000X, D8000X, D9000X, D9200X, E1000X, and G1000X.

Structurally the field is a simple rollover anticlinal structure with a 4 ways dip closed structural style (Figure 2.2). While stratigraphically the field is a channel dominated shallow marine environment, characterised dominantly by tidal channel, Shoreface deposits and heterolithic. These channels are punctuated by lot of baffles and where shale are competent cannot be correlated across (Figure 2.6).

The reservoirs in Egbema Field are described as a series of stacked fluvio-marine sediments laterally correlatable and mostly continuous across the field as shown by the dip cross section (Figure 2.5).

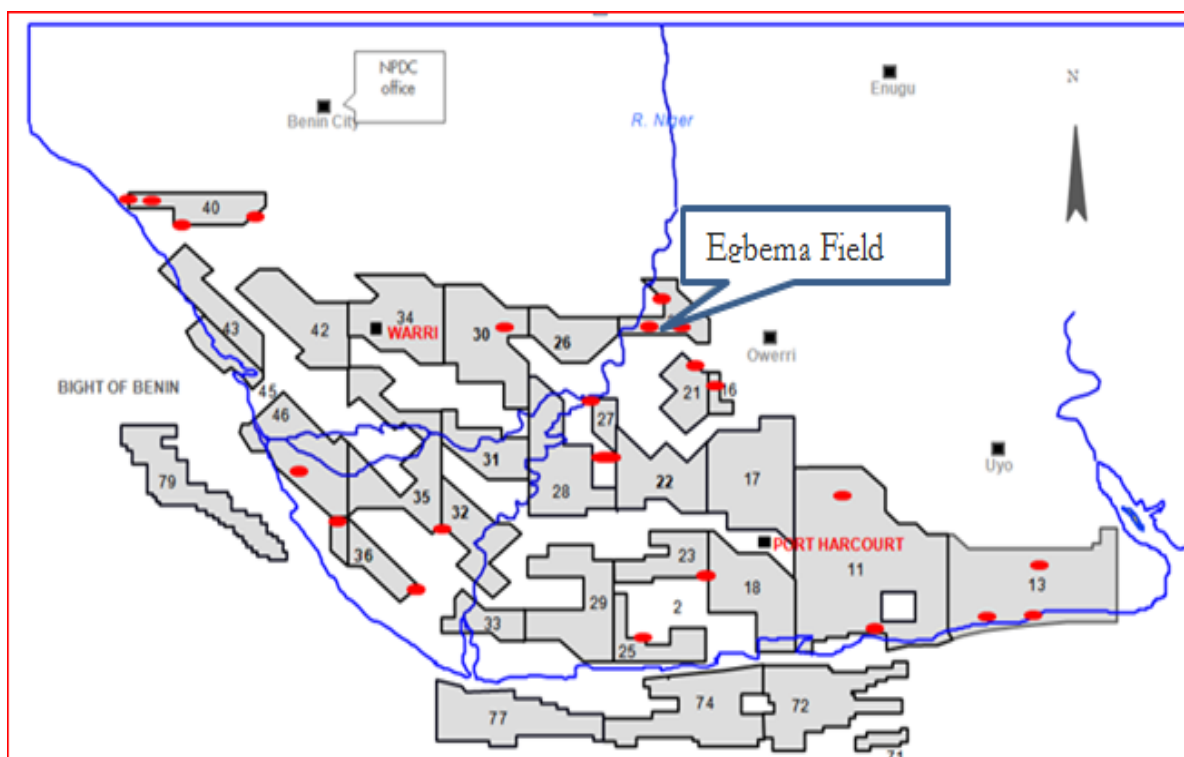


Figure 2.1: Egbema Field Location

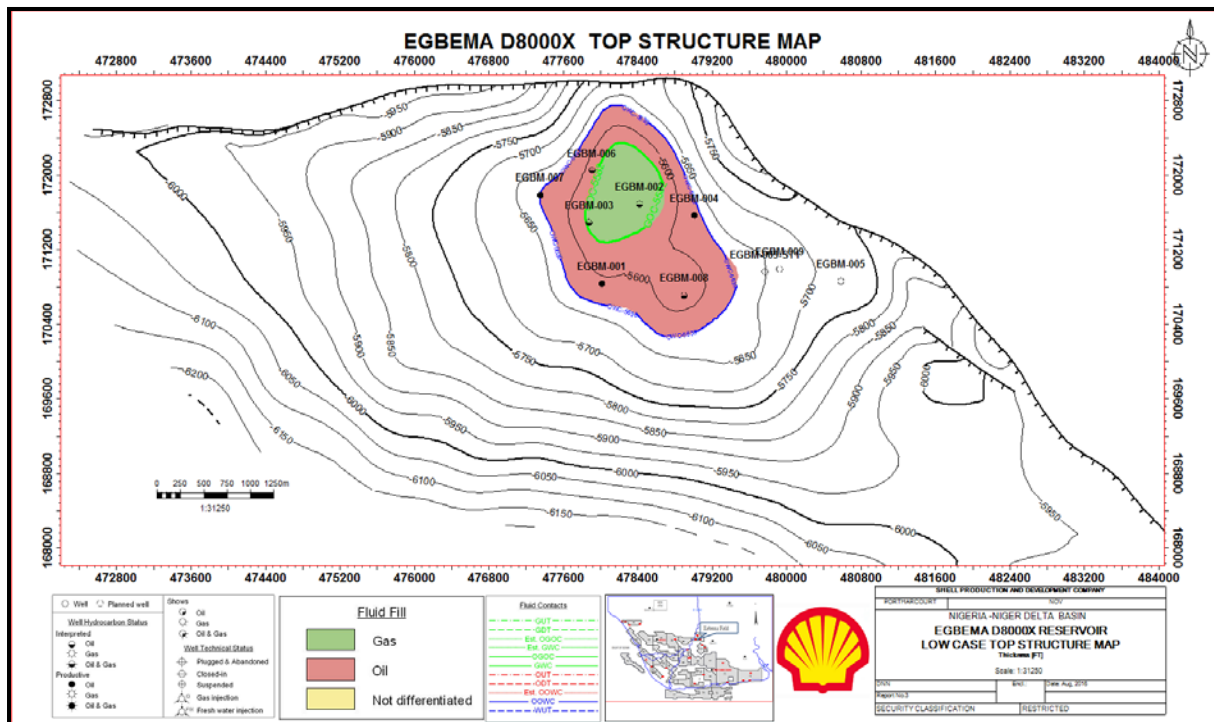


Figure 2.2: Egbema D8000X Top Structure Map

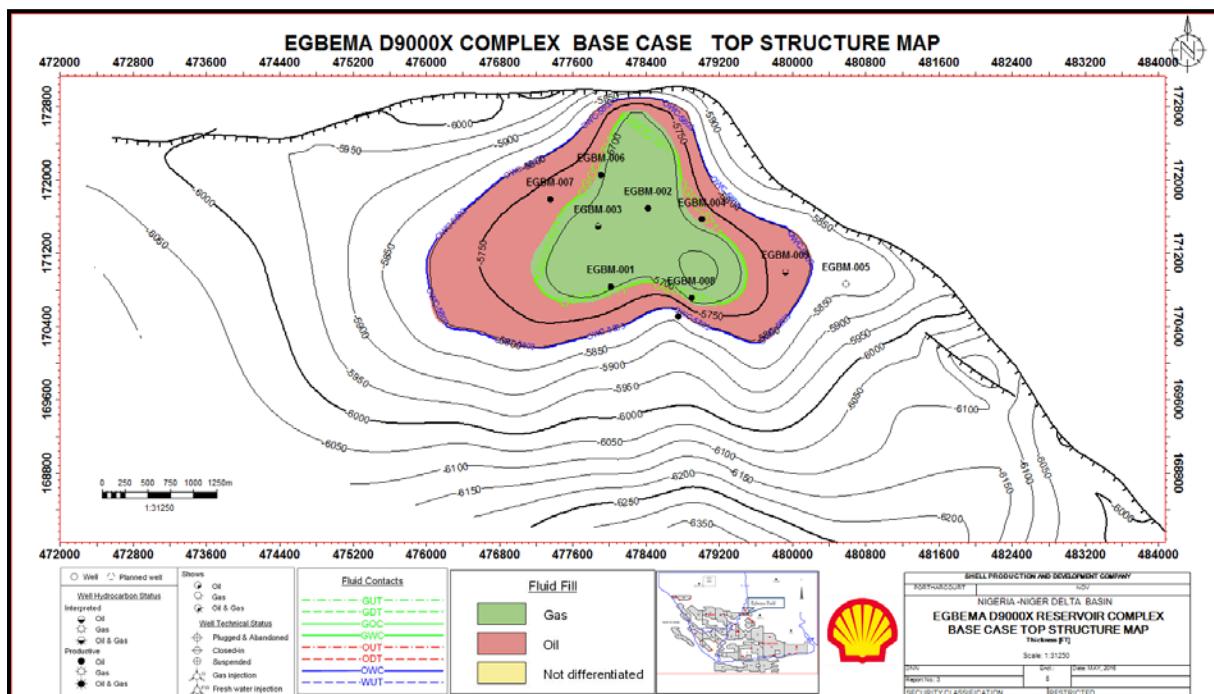


Figure 2.3: Egbema D9000X Complex Top Structure Map

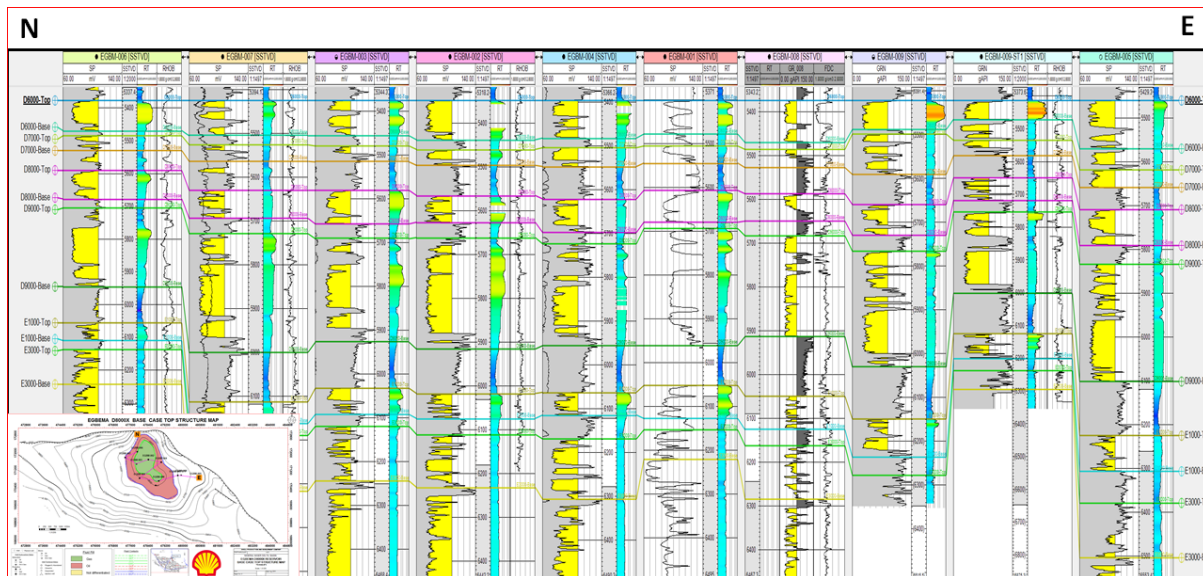


Figure 2.6: Egbema Field Reservoir Correlation

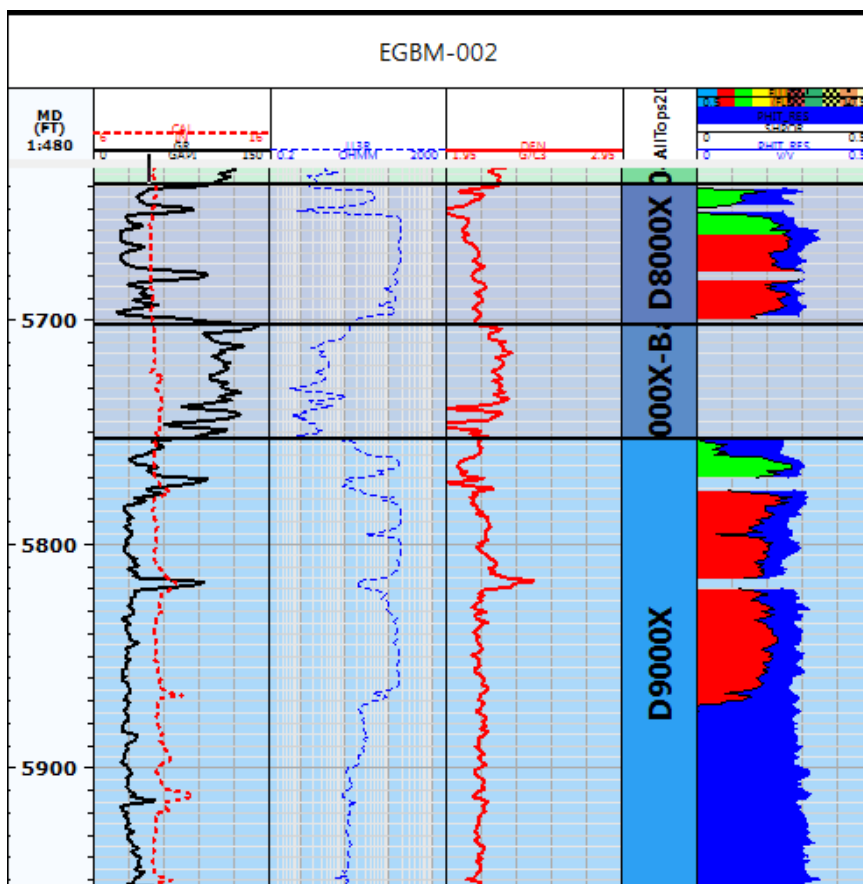


Figure 2.7: Type Log of Egbema-002 across D8000 and D9000 Reservoir complex

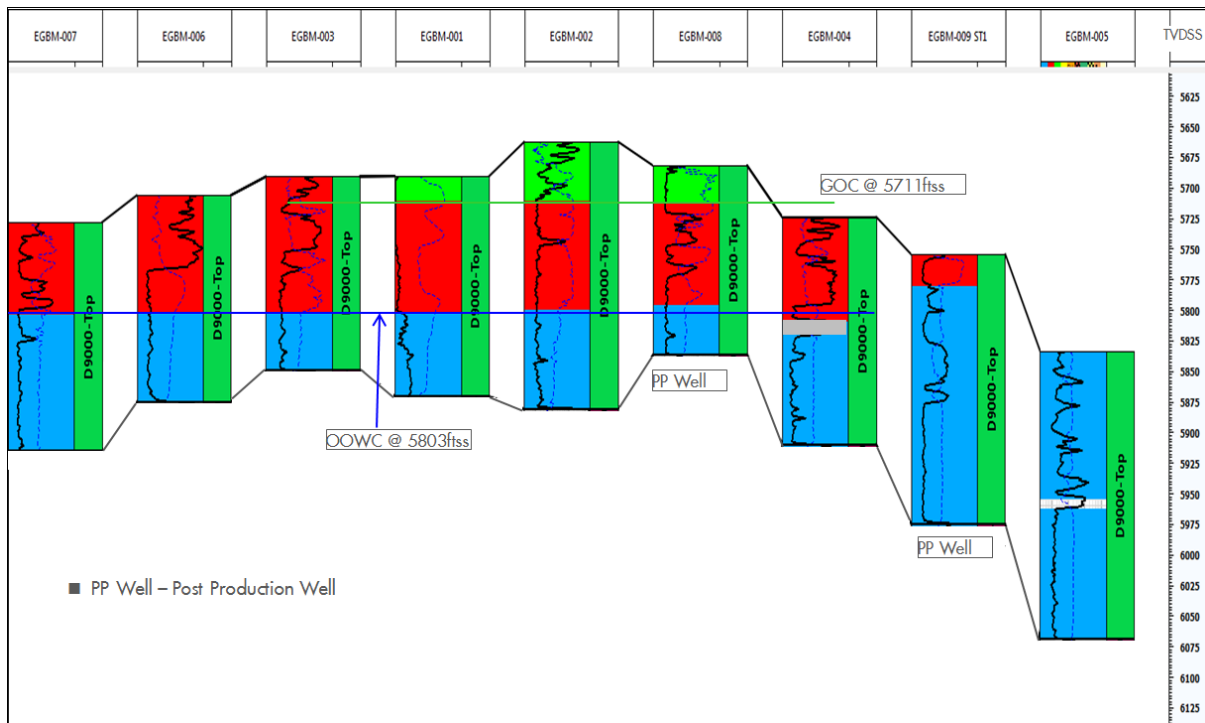


Figure 2.8: Hydrocarbon Distribution Plot for D9000 Reservoir

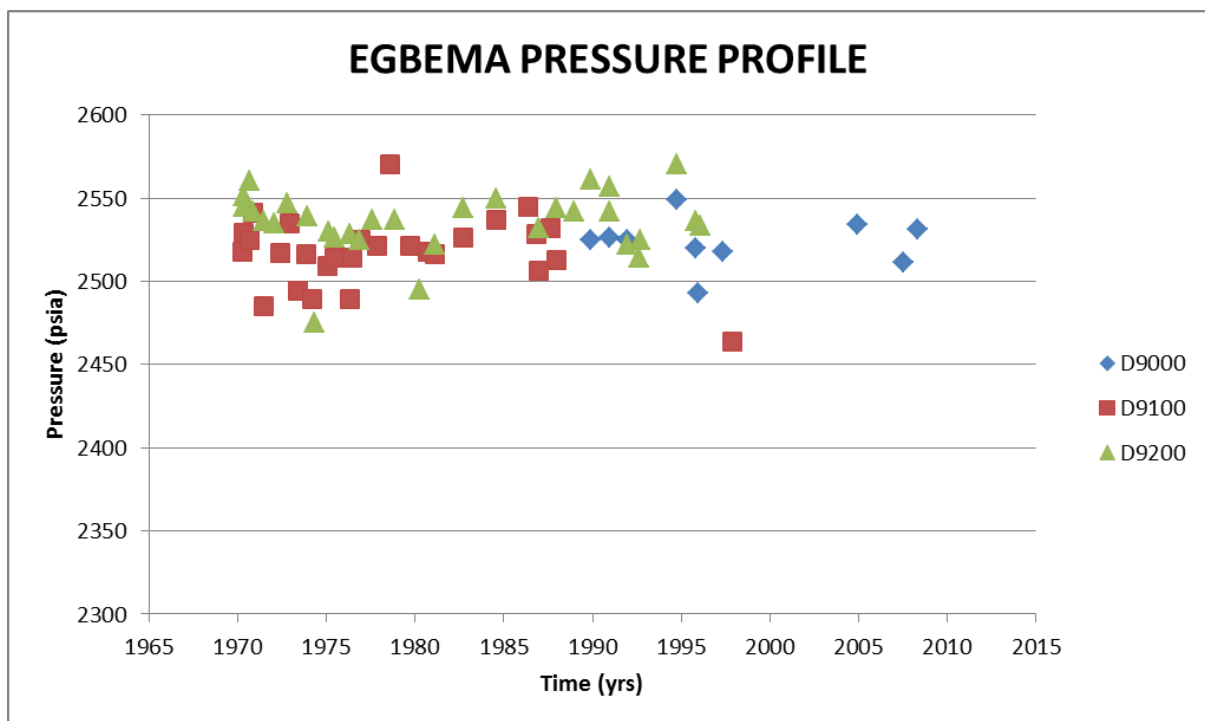


Figure 2.9: D9000X, D91000X & D92000X Historical pressure plot

Egbema field history is summarized in Table 2.1 below.

Table 3.1: Egbema Field Description

Discovery/ On Stream Date	1960/1970
Category/OML	Land/OML 20
AoO (Contract Type)	Tax Royalty (JV)
No. of Active Conduits for Oil	3
Structure	Simple rollover anticline
Depositional Environment	Tidal channel dominated shallow marine
Connectivity	laterally Continuous
Rock Properties	Good ($\phi = 15 - 30 \%$), $K = 800 - 3000 \text{ mD}$)
Oil/Gas Properties	Light Oil/Associated gas
Main Reservoir Energy	Active Water drive
Pressure depletion ($((P_i - P)/P_i) \times 100 \%$)	Negligible
Total No. of Wells/DPs	9/17
Date last well Drilled	1992
No. Of Active Conduits for Gas	0
Number of Reservoirs	12
Well Type	Conventional
Completion Type	IGP/ESS/SCON
Evacuation Facility	Egbema Flow Station
Key Operational Challenge	Wax deposition and high water cut
Peak Production Rate/date	18 Mbopd (September 1972)
No. Of Producing wells/Drainage points	3/3
Current Oil Production Rate (30.09.2016)	783 bopd
Current Gas Production Rate (30.09.2016)	0.87 MMscf/d
No. Of planned well/Type	None
STOIIP/UR, FGIIIP/UR (1/1/2016)	213/100 MMstb, 285/192 Bscf
Current NGL (Condensate) Production Rate	Nil
Cum. Oil Production 30.09.2016	77.1 MMstb
Cum. Gas Production 30.09.2016	55.6 Bscf
Cum. NGL (condensate) Production 31.10.2016	Nil

2.2. Field History

Following discovery in October 1960, Egbema field development commenced and by mid-1967, seven wells had been drilled. The eighth well was drilled in September 1982, following a field review carried out in 1978. The 1987 field review proposed a ninth well which was drilled in 1992. The field started production in April 1970, and achieved a peak of 18,000 bopd in 1972 and has since been in decline with current (September 2016) production of 783 bopd at 64% water cut and GOR of 1,100 scf/bbl from 3 drainage points on different reservoirs namely; D1000X, D8000X, and E1000X.

In 2010, the operatorship of the field (along with other fields in Egbema Node) was transferred to the Nigerian Petroleum Development Company (NPDC). The production performance plot is shown in Fig. 2.10.

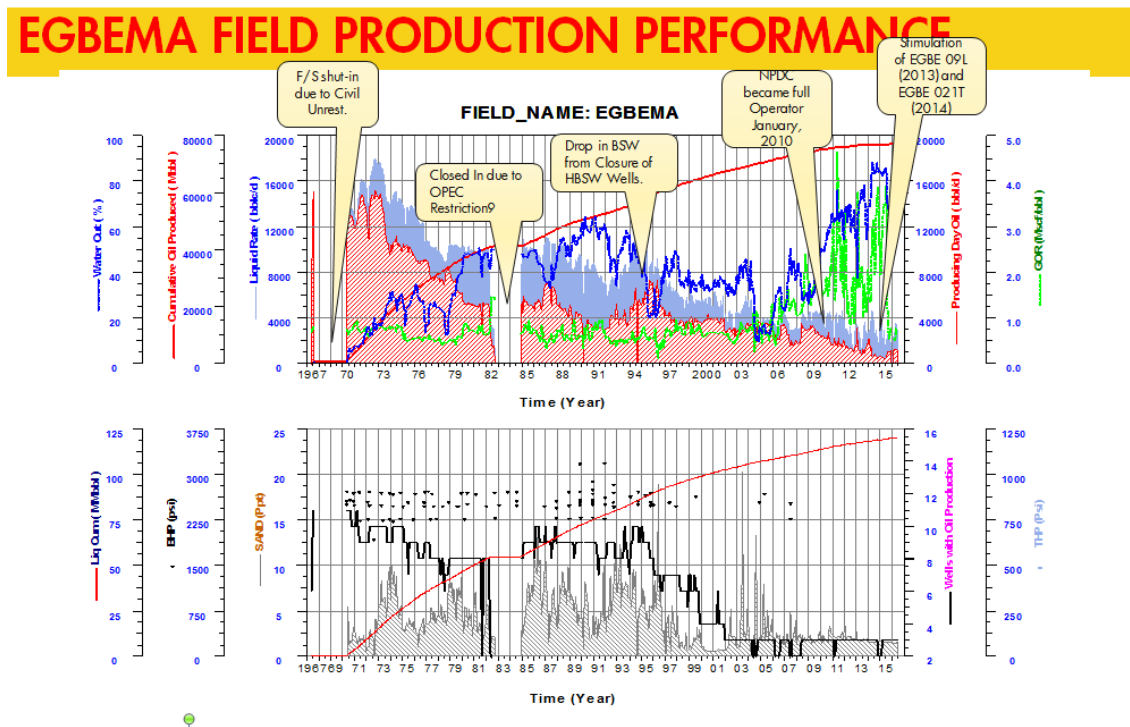


Figure 2.10: Egbema Field Historical Performance Plot

2.3. Field Development Plan Overview

Egbema field was discovered by Egbm-1 well in 1960 and by mid-1967, seven wells had been drilled. The eighth well was drilled in September 1982, following a field review carried out in 1978. The 1987 field review proposed a ninth well which was drilled in 1992.

Currently the operator of Egbema has proposed drilling of infill wells amongst other optimization and restoration opportunities. These opportunities are being matured and will form the basis of the next development phase of the field.

3. ULTIMATE RECOVERY CHANGES

3.1. 1/1/2017 ARPR Updates

Based on seismic reprocessing, re-interpretation and geological model update, PVT re-evaluation, Petrophysics properties and technical limit recovery factor re-evaluation, and analysis of historical production performance, updates have been made to the initially in-place volumes and ultimate recoveries for three reservoirs in Egbema field for 1/1/2017 as shown in Appendix A.1.5.

3.2. Reasons for changes

The reasons for change in the Initially In place and Ultimate Recovery volumes in the affected reservoirs from 1/1/2016 to 1/1/2017 include the following:

- Re-processing and interpretation of seismic data
- Static geological modeling and volume re-estimation
- Re-evaluation of petrophysical properties
- Re-evaluation of PVT parameters
- Review of technical limit recovery factors based on observed production performance

3.2.1. *Re-processing and interpretation of seismic data*

3D seismic was acquired across OML 20 in 1994 and processed to Post-Stack Time Migration in 1995 to 1996 by SPDC SIPMAP and Pre-stack Time Migration in 2009. Further re-processing of the 3D seismic was carried out in 2013 to PSDM (Post –Stack Depth Migration) by IDSL for the current studies. This reprocessed Post-Stack Depth Migration (PSDM) seismic data was transferred to SPDC for a re-interpretation of the project target Reservoirs (D8000X, D9000X, complex and E1000X) to validate the interpretation carried out by IDSL on behalf of NPDC. The horizon interpretation by SPDC is quite comparable to that of NPDC for the reservoirs, except in the E1000X reservoir where there was some depth difference in the loop interpreted. SPDC did not carry out a full field Seismic interpretation. The SPDC interpretation of the candidate reservoirs provided the basis for the current field evaluation and development premises.

Depth uncertainties were evaluated for the three reservoirs ranges from (+/-52 ft) for D8000X, D9000X (+/- 41 ft) and E1000X (+/- 78 ft).

3.2.2. *Static geological modelling and volume re-estimation*

Field wide correlation was carried out for the Egbema main, (Figure 2.6). This correlation was done by picking the well tops and bases and interpretation of log signatures, after which they were reconciled with that of the Petrophysicist. The principles of stratigraphy, conceptual geological model and log signatures were the basis for depositional environment, reservoir architecture and facies type's interpretations. The environment of deposition is interpreted to be tidal channel dominated shallow marine environment, characterised dominantly by tidal channel, Shoreface deposits, shales and heterolithic.

To establish the static connectivity of the D8000X, D9000X Complex and E1000 reservoirs, correlation was carried out across the 10 wells penetrating the reservoir, the D8000X and E1000X reservoirs were reliably correlated while, the D9000X Complex reservoir could not be, due to the presence of the thin shale baffle and where there are competent thick shale they do not act as barriers to flow, this is also confirmed by similar historical pressure (Figure 2.9) and original fluid contact.

3.2.3. Re-evaluation of petrophysical properties

A new petrophysical evaluation was carried out to validate NPDC reported properties. The work scope covered the detailed review of sand tops/bases, porosity and saturation determination, fluid typing and contacts analyses.

Petrophysical data availability table, tables of sum and averages and contact details are provided in Appendix A1.2. Table 3-1 below shows the methodology for the petrophysical parameters.

Table 3.1: Table of Petrophysical Evaluation

Property	Logs/data used	Methodology / Assumptions	Reservoirs
Reservoir Type	Gamma ray, spontaneous potential, resistivity logs	Primarily determined based on Gamma Ray log signature but supported by other logs.	D8000X, D9000X, E1000X
Hydrocarbon Interval	Resistivity	Deep Resistivity log	D8000X, D9000X, E1000X
Hydrocarbon typing	Production data, logs and Sidewall Sample fluorescence tests interpretation	Hydrocarbon typing based on SWS data	D8000X, D9000X, E1000X
Contact	Resistivity, Sidewall Sample fluorescence tests interpretation	Fluid contact was based on resistivity log response	D8000X, D9000X, E1000X
Porosity	Density and Gamma Ray	Deterministic, matrix/fluid density 2.65/1.0/cc. Porosity-Gamma ray transforms	D8000X, D9000X, E1000X
Net / Gross	Gamma Ray	Shale volume cut-off applied based on outcome of HCPORTH sensitivity on each reservoir	D8000X, D9000X, E1000X
HC Saturation	Resistivity, Density, Gamma ray, etc.	Juhasz Normalized Qv Shaly Sand model.	D8000X, D9000X, E1000X

3.2.4. Re-evaluation of PVT parameters

PVT properties for the D8000, D9000 Complex and E1000 reservoirs were re-evaluated using a combination of Niger Delta correlation, standing correlation and Egbema field correlation to model and capture the uncertainty range. The standing correlation was used for the base case as established in 1987 study of Egbema field (Ref 3). The CGRi estimates were based on the Seiders' correlation.

Table 3.2: Changes in PVT Property

Reservoir	Parameter	Unit	Present	Previous
D8000X	Initial Pressure	Psia	2500	2500
	Bubble Point Pressure	Psia	2500	2473
	RSI	scf/bbl	321	443
	Boi	rbbl/stb	1.129	1.174
	CGRi	rbbls/MMscf	2.644	
	Bgi	rbbl/Mscf	1.12	1.072
	Res Temp	Deg. F	162	158
	Specific Gravity		0.92	0.92
	Oil Viscosity	cP	0.4	0.4
	Water Viscosity	cP	2.59	2.59
D9000 Complex	Initial Pressure	Psia	2550	2550
	Bubble Point Pressure	Psia	2550	2550
	RSI	scf/bbl	389	423
	Boi	rbbl/stb	1.192	1.199
	CGRi	rbbls/MMscf	4.191	
	Bgi	rbbl/Mscf	1.11	1.047
	Res Temp	Deg. F	164	158
	Specific Gravity		0.89	0.89
	Oil Viscosity	cP	0.4	0.4
	Water Viscosity	cP	1.54	1.54
E1000X	Initial Pressure	Psia	2725	2725
	Bubble Point Pressure	Psia	2725	2725
	RSI	scf/bbl	588	626
	Boi	rbbl/stb	1.29	1.379
	CGRi	rbbls/MMscf	9.46	
	Bgi	rbbl/Mscf	1.04	0.995
	Res Temp	Deg. F	170	165
	Specific Gravity		0.84	0.81
	Oil Viscosity	cP	0.38	0.38
	Water Viscosity	cP	0.43	0.43

3.2.5. Review of technical limit recovery factors based on observed production performance

Technical limit recovery factor for 1/1/2017 were re-evaluated analytically using the program (SUITE) that uses Monte Carlo technique. This was done to respect the newly evaluated parameters which are inputs into recovery factor evaluation.

3.3. Methodology and Results

The technical ultimate recoveries were volumetrically estimated after applying the technical limit recovery factors on the in place volumes

The details of the evaluations for each reservoir are presented on Appendix A1.1.

3.3.1. Uncertainty Assessment

The key static/dynamic uncertainties that influence reserves estimates for the Egbema field are discussed below. The impact of these uncertainties for the key reservoirs is illustrated in the Tornado Plots shown in Figure 3-1 to 3-3.

- Structure

Structural uncertainty as reflected by the GRV is one of the uncertainties associated with Egbema reservoirs, these has an impact on the volumes. Most of the wells are placed crestally, with uncertainty at the flank. The structural/depth uncertainty increases away from the area of well penetration towards the flanks of the reservoir. To manage this uncertainty and provide better structural control for estimating gross reservoir rock volume, three realisations were considered for all the reservoirs based on seismic depth uncertainty of +/-52, +/-41/, +/-78 ftss for the D8000X, D9000X Complex and E1000X reservoirs relative to the base case map respectively. The realisations corresponding to the Low, and High case structures were used to capture the uncertainty band in the reservoirs.

- Porosity

Porosity was evaluated as total Porosity from density in wells with density log (EGM wells 002, 006, 007 and 008). A Gamma ray-porosity transform was used to estimate synthetic porosity in wells without density data. Same was applied in poor density data/washed out intervals. Average reservoir porosity was determined from the average properties estimated in all wells. Uncertainty range around the porosity was determined using probabilistic methods.

- Hydrocarbon saturation

Hydrocarbon saturation was evaluated using Archie (clean sand) and Normalized Qv (Juhash-shaly sand) models. Saturation was computed for all Egbema Wells using both methodologies. Result obtained from both was found to match nicely in the clean sand intervals. However, in the shaly intervals, Juhasz which is a modified form of Archie accounts for added clay conductivity. Juhasz model was found to be more representative; hence used as final saturation model. The D8000X reservoir is oil bearing in six penetrations (EGBM- 006, 003, 001, 002, 008, and 004) with gas penetrated in wells EGBM-003, 002 and wet in wells EGBM-005, 007 & 009. D9000 reservoir complex is oil bearing in nine penetrations (EGBM-001, 002, 003, 004, 006, 007, 008, 009 & 009ST) with gas penetrated in wells EGBM-001, 002, 003 and wet in EGBM-005. E1000X reservoir is oil bearing in six penetrations (EGM-001, 004, 007, 008, 009 & 009ST) with gas penetrated in wells EGBM-001, 002, 004, 003, 006, 008 and wet in EGM-005. Average oil and gas saturations have been estimated using well averages. Uncertainty ranges were estimated using probabilistic methods.

- Fluid Contacts

Sidewall samples (sws), resistivity and Neutron/density log data and production data, were used for the fluid interpretation and contact analyses. Hydrocarbon typing was based on sidewall sample fluorescence test results. D8000 reservoir oil-water contact was logged in EGBM -001, 003 & 006 wells. Low and high case contacts are same as base case. Gas-oil contact is based on sws; shallowest and deepest samples have been used as low/base and high case contacts respectively. D9000 reservoir oil-water contact was logged in EGBM-001, 002 and 006 wells. Low and high case contacts are same as base contact. Gas-oil contact is also been established from sws in EGBM-008 well. E1000 reservoir oil-water contact on the other hand was estimated based on hydrocarbon column equivalent to oil produced between June 1967 (onset of production) and April 1992 when EGBM-009 well was drilled. E1000 Gas-oil contact was established from sws in EGM-001 and 008 wells.

- Sums and Averages

For the pay zones of each well, NTG, Average Porosity and Average Water saturation were computed and given in the Tables below

- Uncertainty analysis and Petrophysical properties inputs to HCIIP calculation

The spread (P90, P50, P10) in the reservoir petrophysical properties (porosity, saturation and net-to-gross) which were input to the volumetric evaluation was generated using the PDO uncertainty analysis package (statistical package that generates volumetric inputs taking into consideration systematic errors, statistical errors and geological variations) Table 3.3 below shows the uncertainty in petrophysical data for the reservoirs.

Table 3.3: Egbema Property ranges

Egbema Property Ranges												
Reservoir	N/G			Por			Sg			So		
	P90	P50	P10	P90	P50	P10	P90	P50	P10	P90	P50	P10
D8000	0.81	0.86	0.90	0.27	0.29	0.30	0.47	0.66	0.84	0.63	0.76	0.89
D9000	0.56	0.64	0.73	0.27	0.29	0.31	0.45	0.57	0.69	0.46	0.57	0.67
E1000	0.69	0.76	0.82	0.24	0.26	0.28	0.59	0.70	0.81	0.50	0.64	0.77

3.3.2. HCIIP Calculations

Hydrocarbon initially in-place for the different reservoirs were calculated using the 'Volume calculation' tool in Petrel. A Tornado plot of the reservoir HCIIP generation parameters or properties (GRV, FVF, NTG, Porosity and Saturation ranges) was generated to deduce their impact on the static volume. The parameters were then plotted and strung on the uncertainty matrix for each reservoir to derive the Low, Base and high case In-place volume realisations for each reservoir based on the Tornado analysis and assumptions.

The low, base and high case reservoir properties combined with the GRV ranges realizations were used to calculate the resultant low, base and high case In-place volume estimates for the D8000X, D9000X complex and E1000X reservoirs (Reference Figures 3.1 to 3.3). Thus the

reservoir parameter realizations were used to generate the P1, P1+P2 and P1+P2+P3 case hydrocarbon initially in-place volume estimate for these reservoirs.

Figure 3.1: Egbema D8000X Uncertainty Matrix

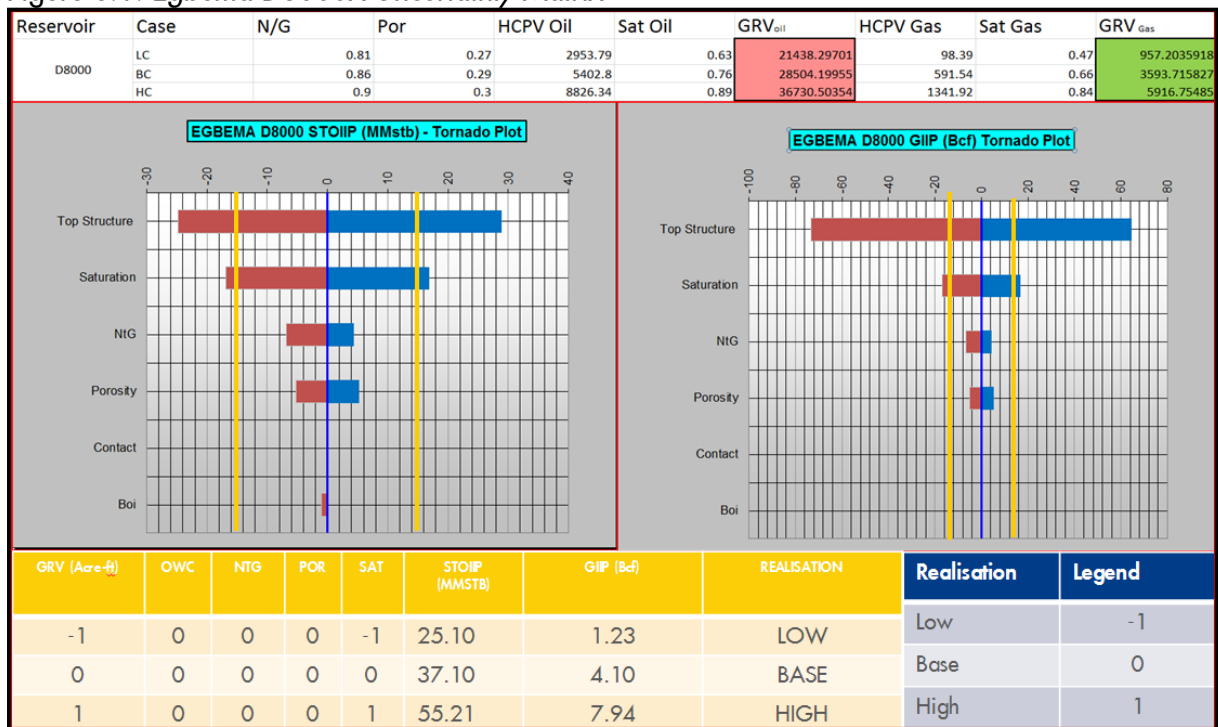


Figure 3.2: Egbema D9000X Reservoir Uncertainty Matrix

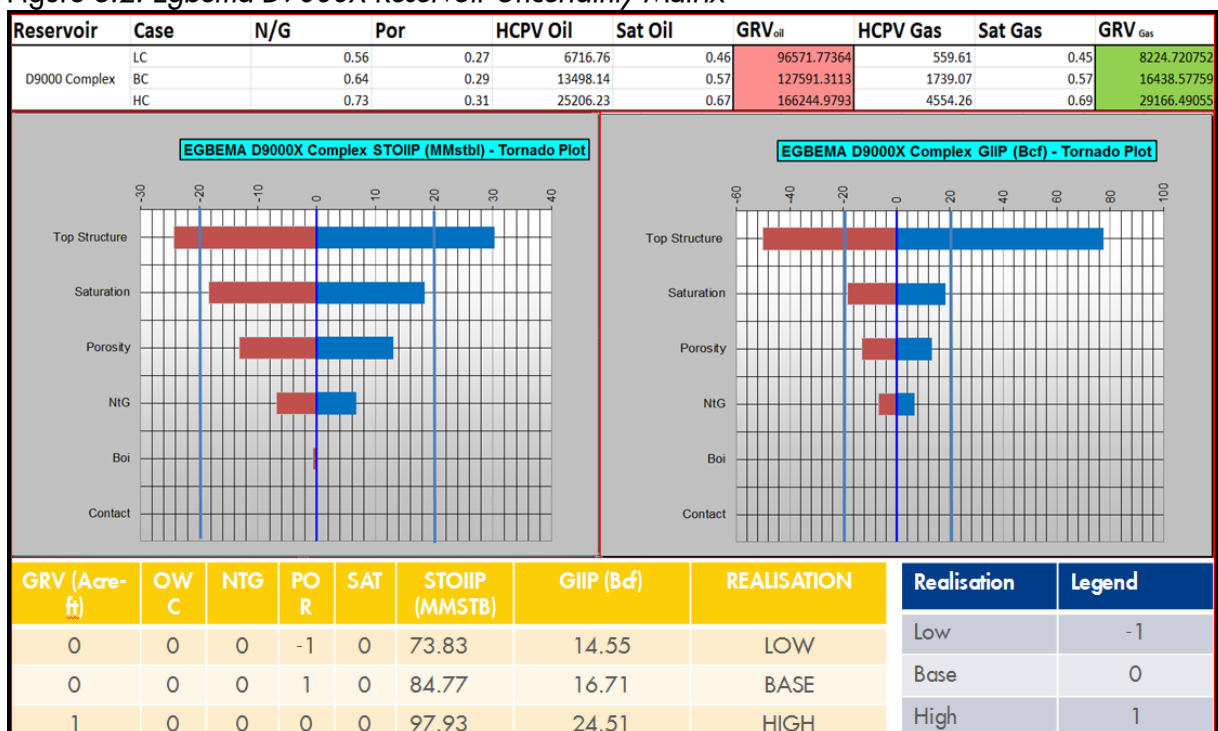


Figure 3.3: Egbema E1000X Reservoir Uncertainty Matrix

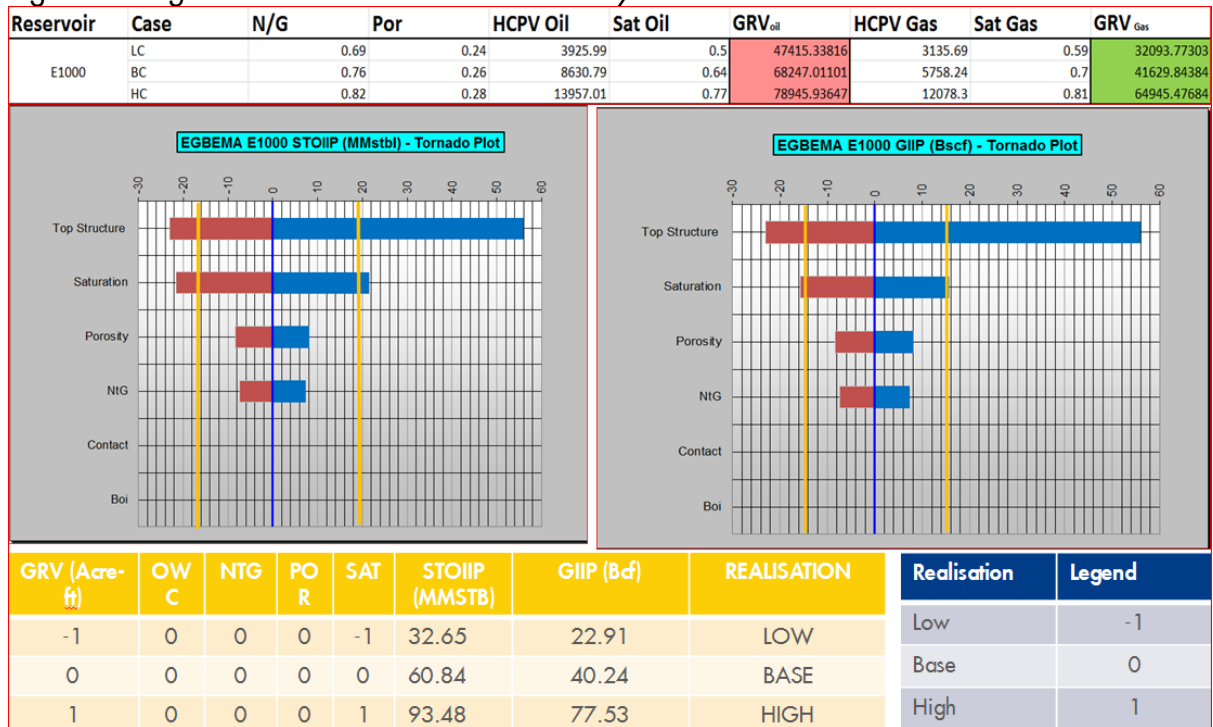


Table 3.4: Summary of Stock Tank Oil Initially in Place (STOIIP) Changes

	01/01/2016 (MMstb)			01/01/2017 (MMstb)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	33.9	33.9	33.9	25.1	37.01	55.21
D9000X	41.9	50.6	60	0	0	0
D9200X	24.6	28.7	33	0	0	0
D9000 Complex	0	0	0	73.83	84.77	97.93
E1000X	24.6	28.1	28.1	32.65	60.84	93.48

Table 3.5: Summary of Solution Gas Initially in Place (SGIIP) Changes

	01/01/2016 (Bscf)			01/01/2017 (Bscf)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	15.0	15.0	15.0	6.93	11.88	20.21
D9000X	16.6	20.2	24.1	0	0	0
D9200X	9.8	11.5	13.3	0	0	0
D9000 Complex	0	0	0	14.77	32.98	56.60
E1000X	15.4	17.6	17.6	19.19	35.77	54.97

Table 3.6: Summary of Gas Cap Gas Initially in Place (GCGIIP) Changes

	01/01/2016 (Bscf)			01/01/2017 (Bscf)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	6.1	6.1	6.1	1.23	4.1	7.94
D9000X	12	14.4	17.1	0	0	0
D9200X	0	0	0	0	0	0
D9000 Complex	0	0	0	14.55	16.71	24.51
E1000X	29	36.8	45.6	27.91	40.24	77.53

Table 3.7: Summary of Non-Associated Gas Initially in Place (NAGIIP) Changes

	01/01/2016 (Bscf)			01/01/2017 (Bscf)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	0	0	0	0	0	0
D9000X	0	0	0	0	0	0
D9200X	0	0	0	0	0	0
D9000 Complex	0	0	0	0	0	0
E1000X	0	0	0	0	0	0

Table 3.8: Summary of Condensate Initially in Place (CIIP) Changes

	01/01/2016 (MMstb)			01/01/2017 (MMstb)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	0	0	0	0.003	0.01	0.02
D9000X	0	0	0.1	0	0	0
D9200X	0	0	0	0	0	0
D9000 Complex	0	0	0	0.06	0.07	0.10
E1000X	0	0	0	0.22	0.38	0.74

3.3.3. Recovery Factor Evaluation

Technical limit recovery factor for 1/1/2017 were evaluated analytically using the program (SUITE) that uses Monte Carlo technique.

The input parameters into SUITE for the estimation of range of recovery factors include:

- (1) Initial reservoir pressure and temperature
- (2) Pressure depletion obtained from MBAL
- (3) Range of PVT parameters obtained from correlation
- (4) Range of petrophysical parameters as obtained from the petrophysical evaluation for Egbema FOD reservoirs.
- (5) Residual fluid residual saturations obtained from RELATE
- (6) Sweep efficiency obtained from SUITE package inbuilt correlation.
- (7) Area versus depth data obtained from updated reservoir geological models.

Appendix A1.1a and A1.1b shows the input data and output for the oil and solution gas and gas cap respectively. Details are in reference 1.

Table 3.9: Uncertainty ranges for Recovery Factor for Oil

	01/01/2016 Recovery Factor (%)			01/01/2017 Recovery Factor (%)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	10	25	25	54	58	62
D9000X	56	50	43	0	0	0
D9200X	58	50	43	0	0	0
D9000 Complex	0	0	0	56	62	68
E1000X	60	60	60	57	63	67

Table 3.10: Uncertainty ranges for Recovery Factor for Solution Gas

	01/01/2016 Recovery Factor (%)			01/01/2017 Recovery Factor (%)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	25	25	25	55	59	63
D9000X	71	58	49	0	0	0
D9200X	46	46	46	0	0	0
D9000 Complex	0	0	0	56	63	69
E1000X	150	220	105	58	64	68

Table 3.11: Uncertainty ranges for Recovery Factor for Free Gas

	01/01/2016 Recovery Factor (%)			01/01/2017 Recovery Factor (%)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	64	66	66	38	48	58
D9000X	47	59	68	0	0	0
D9200X	0	0	0	0	0	0
D9000 Complex	0	0	0	33	45	56
E1000X	70	70	70	32	42	52

Table 3.12: Uncertainty ranges for Recovery Factor for Condensate

	01/01/2016 Recovery Factor (%)			01/01/2017 Recovery Factor (%)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	0	0	0	37	45	54
D9000X	0	0	0	0	0	0
D9200X	0	0	0	0	0	0
D9000 Complex	0	0	0	33	43	53
E1000X	0	0	0	32	42	52

3.3.4. Ultimate Recovery Evaluation

The ultimate recovery of the reservoirs was evaluated by applying the recovery factors estimated as described in Section 3.3.3 to the HCIIPs.

Table 3.13: Summary of Oil Ultimate Recovery Ranges

	01/01/2016 Ultimate Recovery (MMstb)			01/01/2017 Ultimate Recovery (MMstb)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	3.5	8.5	8.5	13.55	21.47	34.23
D9000X	23.4	25.7	25.7	0	0	0
D9200X	14.3	14.3	14.3	0	0	0
D9000 Complex	0	0	0	41.34	52.56	66.59
E1000X	14.8	16.9	16.9	18.61	38.33	62.63

Table 3.14: Summary of Solution Gas Ultimate Recovery Ranges

	01/01/2016 Ultimate Recovery (Bscf)			01/01/2017 Ultimate Recovery (Bscf)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	3.8	3.8	3.8	3.81	7.01	12.73
D9000X	11.7	11.7	11.7	0	0	0
D9200X	4.5	5.3	6.1	0	0	0
D9000 Complex	0	0	0	8.27	20.77	39.06
E1000X	23.3	39.5	46.5	11.13	22.90	37.38

Table 3.15: Summary of Free Gas Ultimate Recovery Ranges

	01/01/2016 Ultimate Recovery (Bscf)			01/01/2017 Ultimate Recovery (Bscf)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	3.9	4	4	0.47	1.97	4.61
D9000X	5.6	8.5	11.7	0	0	0
D9200X	0	0	0	0	0	0
D9000 Complex	0	0	0	4.80	7.52	13.73
E1000X	20.3	25.8	32	8.93	16.90	40.32

Table 3.16: Summary of Condensate Ultimate Recovery Ranges

	01/01/2016 Ultimate Recovery (MMstb)			01/01/2017 Ultimate Recovery (MMstb)		
	Low Case	Base Case	High Case	Low Case	Base Case	High Case
D8000X	0	0	0	0.001	0.005	0.011
D9000X	0	0	0	0	0	0
D9200X	0	0	0	0	0	0
D9000 Complex	0	0	0	0.02	0.03	0.05
E1000X	0	0	0	0.09	0.16	0.38

3.3.5. Future Development Plans

Currently the operator of Egbema has proposed drilling of infill wells amongst other optimization and restoration opportunities. These opportunities are being matured and will form the basis of the next development phase of the field.

References

1. Egbema FOD Reservoirs Recovery Factor Evaluation. November 2016
2. Egbema Main Field HCIIP Volume Report. November 2016.

APPENDIX:**A1.1. Methodology for Recovery Estimation Table**

Reservoir	Low Estimate	Best Estimate	High Estimate	Comments
D8000X	Volumetrics/ Analytical techniques	Volumetrics/ Analytical techniques	Volumetrics/ Analytical techniques	Low-Base-High technical limit RF estimated via the program (SUITE) that uses Monte Carlo techniques
D9000 Complex	Volumetrics/ Analytical techniques	Volumetrics/ Analytical techniques	Volumetrics/ Analytical techniques	Low-Base-High technical limit RF estimated via the program (SUITE) that uses Monte Carlo techniques
E1000	Volumetrics/ Analytical techniques	Volumetrics/ Analytical techniques	Volumetrics/ Analytical techniques	Low-Base-High technical limit RF estimated via the program (SUITE) that uses Monte Carlo techniques

A1.1a. SUITE results sheet for oil and solution gas range of recovery factors

REC-FACTORY ver 1.0																		
Reservoir		Oil_Col	Ha	mu	Boi	Pb	Rsi	Ew	Eg	Scw	Sorw	Sorg	Sgr	Sgc	DPPI	Rp	RFo	Rfsg
egbema,d8000	minimum	43	10	0.4	1.11	2500	276	0.88	0.69	0.11	0.14	0.16	0.18	0	0.01	276	0.54	0.55
	most likely	43	10	0.4	1.13	2500	321	0.88	0.69	0.24	0.15	0.29	0.19	0.01	0.06	321	0.58	0.59
	maximum	43	10	0.4	1.16	2500	366	0.88	0.69	0.37	0.15	0.42	0.19	0.01	0.1	366	0.62	0.63
egbema,d9000	minimum	92	6	0.4	1.14	2550	200	0.88	0.69	0.33	0.14	0.38	0.16	0	0.01	200	0.56	0.56
	most likely	92	6	0.4	1.19	2550	389	0.88	0.69	0.43	0.15	0.48	0.18	0.01	0.06	389	0.62	0.63
	maximum	92	6	0.4	1.2	2550	578	0.88	0.69	0.54	0.15	0.59	0.19	0.01	0.1	578	0.68	0.69
egbema,e1000	minimum	80	8	0.38	1.28	2725	588	0.88	0.69	0.23	0.15	0.28	0.17	0	0.01	588	0.57	0.58
	most likely	80	8	0.38	1.29	2725	588	0.88	0.69	0.36	0.15	0.41	0.18	0.02	0.06	588	0.63	0.64
	maximum	80	8	0.38	1.3	2725	588	0.88	0.69	0.5	0.15	0.55	0.19	0.02	0.1	588	0.67	0.68

A1.1b. SUITE results sheet for free gas and condensate range of recovery factors

Reservoir		Pres	Ti	gamma	CGR	Pd	Sgi	Sgrw	Egw	Pa_gas	RFfg	RFcond
egbema,d8000	minimum	2500	162	0.57	2.64	2500	0.63	0.18	0.5	2250	0.38	0.37
	most likely	2500	162	0.57	2.64	2500	0.76	0.19	0.6	2359	0.48	0.45
	maximum	2500	162	0.57	2.64	2500	0.89	0.19	0.7	2468	0.58	0.54
egbema,d9000	minimum	2550	164	0.57	4.19	2500	0.46	0.16	0.5	2295	0.33	0.33
	most likely	2550	164	0.57	4.19	2500	0.57	0.18	0.6	2404	0.45	0.43
	maximum	2550	164	0.57	4.19	2500	0.67	0.19	0.7	2513	0.56	0.53
egbema,e1000	minimum	2725	170	0.59	9.46	2725	0.5	0.17	0.5	2724	0.32	0.32
	most likely	2725	170	0.59	9.46	2725	0.64	0.18	0.6	2724	0.42	0.42
	maximum	2725	170	0.59	9.46	2725	0.77	0.19	0.7	2724	0.52	0.52

A1.2. Petrophysical data

Data Availability table for Egbema Field

WELL	GR	SP	CAL	DENS	NEU	SONIC	RES			SWS	SAMPLE	PRESSURE
							LN	LL3	LLD			
EGBEMA-001	✓	✓	✓			✓	✓			✓		
EGBEMA-002	✓	✓	✓	✓			✓	✓		✓		
EGBEMA-003	✓	✓	✓				✓	✓		✓		
EGBEMA-004	✓	✓	✓				✓	✓				
EGBEMA-005	✓	✓	✓				✓	✓				
EGBEMA-006	✓	✓	✓	✓			✓	✓				
EGBEMA-007	✓	✓	✓	✓			✓	✓				
EGBEMA-008	✓	✓	✓	✓	✓				✓			
EGBEMA-009	✓	✓	✓		✓	✓			✓			
EGBEMA-009ST	✓	✓	✓		✓	✓			✓			

Petrophysical averages for Egbema reservoirs

D8000X Oil

Well	Zones	Fluid	Top (ftah)	Bottom (ftah)	Top (ftss)	Bottom (ftss)	Gross (ft)	Net (ft)	N/G (frac)	POR (v/v)	SW (frac)
EGBM-001	D8000	OIL	5685	5713	5602	5630	28	19	0.70	0.26	0.28
EGBM-002	D8000	OIL	5661	5702	5587	5628	41	35	0.86	0.30	0.21
EGBM-003	D8000	OIL	5670	5704	5596	5630	34	34	1.00	0.30	0.13
EGBM-004	D8000	OIL	5696	5703	5621	5628	6	5	0.77	0.28	0.70
EGBM-006	D8000	OIL	5664	5708	5590	5633	43	38	0.88	0.28	0.52
EGBM-008	D8000	OIL	5669	5713	5586	5629	43	37	0.85	0.29	0.22
New Average								28	0.86	0.29	0.26
Previous average									-	0.30	0.26

D8000 Gas

Well	Zones	Fluid	Top (ftah)	Bottom (ftah)	Top (ftss)	Bottom (ftss)	Gross (ft)	Net (ft)	N/G (frac)	POR (v/v)	SW (frac)
EGBM-002	D8000	GAS	5639	5661	5565	5587	22	18	0.82	0.27	0.45
EGBM-003	D8000	GAS	5654	5670	5580	5592	12	10	0.83	0.30	0.19
New Average								14	0.83	0.28	0.32
Previous average									-	-	-

D9000 Oil

Well	Zones	Fluid	Top (ftah)	Bottom (ftah)	Top (ftss)	Bottom (ftss)	Gross (ft)	Net (ft)	N/G (frac)	POR (v/v)	SW (frac)
EGBM-001	D9000	OIL	5794	5886	5711	5803	92	50	0.54	0.28	0.52
EGBM-002	D9000	OIL	5777	5873	5703	5799	96	94	0.97	0.28	0.34
EGBM-003	D9000	OIL	5764	5877	5690	5803	112	66	0.59	0.30	0.29
EGBM-004	D9000	OIL	5798	5884	5724	5809	85	33	0.39	0.28	0.76
EGBM-006	D9000	OIL	5781	5877	5706	5803	96	41	0.43	0.29	0.58
EGBM-007	D9000	OIL	5804	5880	5728	5804	75	55	0.73	0.33	0.59
EGBM-008	D9000	OIL	5795	5879	5712	5795	83	57	0.68	0.29	0.45
EGBM-009 ST1	D9000	OIL	5941	5968	5754	5780	25	22	0.88	0.27	0.29
New Average								52	0.65	0.29	0.48
Previous average									-	-	-

D9000 Gas

Well	Zones	Fluid	Top (ftah)	Bottom (ftah)	Top (ftss)	Bottom (ftss)	Gross (ft)	Net (ft)	N/G (frac)	POR (v/v)	SW (frac)
EGBM-001	D9000	GAS	5777	5794	5694	5711	17	16	0.94	0.29	0.40
EGBM-002	D9000	GAS	5736	5777	5662	5703	41	23	0.56	0.29	0.58
EGBM-008	D9000	GAS	5765	5795	5682	5712	30	29	0.97	0.28	0.33
New Average								23	0.82	0.29	0.57
Previous average									-	-	-

E1000 OIL

Well	Zones	Fluid	Top (ftah)	Bottom (ftah)	Top (ftss)	Bottom (ftss)	Gross (ft)	Net (ft)	N/G (frac)	POR (v/v)	SW (frac)
EGBM-001	E1000	OIL	6109	6124	6108	6123	15	12	0.79	0.26	0.46
EGBM-004	E1000	OIL	6108	6115	6108	6115	7	4	0.50	0.28	0.37
EGBM-007	E1000	OIL	6112	6163	6112	6163	51	42	0.82	0.24	0.67
EGBM-008	E1000	OIL	6104	6125	6104	6125	21	12	0.57	0.26	0.24
EGBM-009ST	E1000	OIL	6121	6172	6335	6391	56	46	0.83	0.26	0.35
New Average								23	0.70	0.26	0.41
Previous average									0.69	0.29	0.29

E1000 GAS

Well	Zones	Fluid	Top (ftah)	Bottom (ftah)	Top (ftss)	Bottom (ftss)	Gross (ft)	Net (ft)	N/G (frac)	POR (v/v)	SW (v/v)
EGBM-001	E1000	GAS	6136	6188	6052	6104	52	51	0.98	0.26	0.23
EGBM-002	E1000	GAS	6095	6149	6021	6075	54	39	0.72	0.25	0.40
EGBM-003	E1000	GAS	6106	6166	6032	6092	60	40	0.67	0.29	0.27
EGBM-004	E1000	GAS	6137	6180	6062	6104	42	40	0.95	0.28	0.31
EGBM-006	E1000	GAS	6131	6178	6056	6105	49	25	0.51	0.25	0.70
EGBM-008	E1000	GAS	6132	6188	6049	6104	55	46	0.84	0.27	0.32
New Average								52	0.78	0.27	0.30
Previous average								-	-	-	-

Contact details for Egbema Reservoirs (Current Review)

Zones	GUT (ftss)	GDT (ftss)	GOC (ftss)	OUT (ftss)	ODT (ftss)	OWC (ftss)	WUT (ftss)
D8000X	-	-	5587	-	-	5630	-
D9000	-	-	5711	-	-	5803	-
E1000X	-	-	6104	-	-	6174	-

A1.3. Summary Table of Properties (NNS ARPR Format)

Table 1A: Oil and Solution Gas Properties

RESERVOIR	OWC/ODT _ft	Press_ psia	GOC/OUT _ft	Press_ psia	PB_psia	RSI_scf/b	Boi_v/v	TR_deg.F	SG	Oil Visc_cp	Water Visc_cp
D8000X	5630	2517	5587	2500	2500	321	1.129	162	0.92	0.4	2.59
D9000 Complex	5803	2585	5711	2550	2550	389	1.192	164	0.89	0.4	1.54
E1000X	6174	2750	6104	2725	2725	588	1.29	170	0.84	0.38	0.43

Table 1B: Free Gas and Condensate Properties

No change

A1.4. Summary Table of Properties (NNS ARPR Format)

Table 1B: Free Gas and Condensate Properties

RESERVOIR	GWC/GDT _ft	Press_ psia	GUT_ft	TR_deg.F	Pdew_psia	Density	Z	EI	Mol Weight	CGRi
D8000X	5587	2500	5562	162	2500	8	0.894	159		2.644
D9000 Complex	5711	2550	5628	164	2550	8	0.896	160		4.191
E1000X	6104	2725	6014	170	2725	9	0.895	171		9.46

Table 1C: Physical Parameters

	Net Oil Sand Volume				Sand/ Shale Factor	Porosity	Water sat	Min oil column	Avg net pay	Net Gas Sand Volume			
	Thousand Acre-Ft									Thousand Acre-Ft			
RESERVOIR	LOW	MED	HIGH	EXP	%	%	%	ft	ft	LOW	MED	HIGH	EXP
D8000X	21.4	28.5	36.73	28.5	86	29	24	43	59	0.96	3.59	5.92	3.59
D9000 Complex	96.57	115.16	199.25	115.16	64	29	43	92	112	8.22	21.14	29.16	21.14
E1000X	47.42	78.0	80.0	78.0	76	26	36	80	129	32.03	41.63	64.95	41.63

A1.5. Volume-Initially-In-Place And Ultimate Recovery Change Tables

Table 2A: Oil in-Place & Ultimate Recovery Changes - NNS P1-P2-P3 Method (MMstb)

Reservoir	Reservoir Changes	Reasons for Change	Reference Document	1/1/2016 ARPR						1/1/2017 ARPR						Change During 2016					
				Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery		
				P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
D8000X	Change	Revision	1/1/2017 URCR Update	33.9	0.0	0.0	3.5	5.0	0.0	25.1	11.9	18.2	13.6	7.9	12.8	-8.8	11.9	18.2	10.1	2.9	12.8
D9000X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	41.9	8.7	9.4	23.4	2.3	-	0	0	0	0	0	0	-41.9	-8.7	-9.4	--23.4	-2.3	-
D9200X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	24.6	4.1	4.3	14.3	-	-	0	0	0	0	0	0	-24.6	-4.1	-4.3	-14.3	-	-
D9000 Complex	Change	Revision	1/1/2017 URCR Update	0	0	0	0	0	0	73.8	10.9	13.2	41.3	11.2	14.0	73.8	10.9	13.2	41.3	11.2	14.0
E1000X	Change	Revision	1/1/2017 URCR Update	24.6	3.5	0.0	14.8	2.1	0.0	32.7	28.2	32.6	18.6	19.7	24.3	8.1	24.7	32.6	3.8	17.6	24.3

Restricted

D1000X	No Change	NA	NA	12.8	0.5	1.0	7.7	0.3	0.6	12.8	0.5	1.0	7.7	0.3	0.6	0.0	0.0	0.0	0.0	0.0	0.0
D4000X	No Change	NA	NA	1.2	0.8	1.3	-	0.5	-	1.2	0.8	1.3	-	0.5	-	0.0	0.0	0.0	0.0	0.0	0.0
D5000X	No Change	NA	NA	1.0	0.7	1.1	-	0.5	-	1.0	0.7	1.1	-	0.5	-	0.0	0.0	0.0	0.0	0.0	0.0
D6000M	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D6000X	No Change	NA	NA	14.7	2.4	2.6	-	-	-	14.7	2.4	2.6	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D7000X	No Change	NA	NA	0.9	0.3	0.7	-	-	-	0.9	0.3	0.7	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D7100X	No Change	NA	NA	2.6	0.2	0.4	-	-	-	2.6	0.2	0.4	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
E3000X	No Change	NA	NA	0.6	0.1	0.3	-	-	-	0.6	0.1	0.3	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
G1000X	No Change	NA	NA	32.9	-	-	23.0	2.6	4.6	32.9	-	-	23.0	2.6	4.6	0.0	0.0	0.0	0.0	0.0	0.0

Table 2B: Solution Gas in-Place & Ultimate Recovery Changes - NNS P1-P2-P3 Method (Bscf)

				1/1/2016 ARPR						1/1/2017 ARPR						Change During 2016					
				Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery		
Reservoir	Reservoir Changes	Reasons for Change	Reference Document	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
D8000X	Change	Revision	1/1/2017 URCR Update	15.0	0.0	0.0	3.8	0.0	0.0	6.9	5.0	8.3	3.8	3.2	5.7	-8.1	5.0	8.3	0.0	3.2	5.7
D9000X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	16.6	3.6	3.9	16.6	3.6	3.9	0	0	0	0	0	0	-16.6	-3.6	-3.9	-16.6	-3.6	-3.9
D9200X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	9.8	1.7	1.8	9.8	1.7	1.8	0	0	0	0	0	0	-9.8	-1.7	-1.8	-9.8	-1.7	-1.8
D9000 Complex	Change	Revision	1/1/2017 URCR Update	0	0	0	0	0	0	14.8	18.2	23.6	8.3	12.5	18.3	14.8	18.2	23.6	8.3	12.5	18.3
E1000X	Change	Revision	1/1/2017 URCR Update	15.4	2.2	0.0	23.3	16.2	7.0	19.2	16.6	19.2	11.1	11.8	14.5	3.8	14.4	19.2	-12.2	-4.4	7.5
D1000X	No Change	NA	NA	2.6	0.1	0.2	3.3	-	-	2.6	0.1	0.2	3.3	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D4000X	No Change	NA	NA													0.0	0.0	0.0	0.0	0.0	0.0

Restricted

				0.3	0.2	0.3	0.1	-	-	0.3	0.2	0.3	0.1	-	-						
D5000X	No Change	NA	NA	0.3	0.2	0.3	0.1	-	-	0.3	0.2	0.3	0.1	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D6000M	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D6000X	No Change	NA	NA	4.2	0.7	0.7	-	-	-	4.2	0.7	0.7	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D7000X	No Change	NA	NA	0.3	-	0.3	-	-	-	0.3	-	0.3	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D7100X	No Change	NA	NA	0.8	-	0.1	-	-	-	0.8	-	0.1	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
E3000X	No Change	NA	NA	0.3	0.1	0.2	-	-	-	0.3	0.1	0.2	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
G1000X	No Change	NA	NA	42.1	-	-	24.6	-	-	42.1	-	-	24.6	-	-	0.0	0.0	0.0	0.0	0.0	0.0

Table 2C: Cap Gas in-Place & Ultimate Recovery Changes - NNS P1-P2-P3 Method (Bscf)

				1/1/2016 ARPR						1/1/2017 ARPR						Change During 2016					
				Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery		
Reservoir	Reservoir Changes	Reasons for Change	Reference Document	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
D8000X	Change	Revision	1/1/2017 URCR Update	6.1	0.0	0.0	3.9	0.1	0.0	1.2	2.9	3.8	0.5	1.5	2.6	-4.9	2.9	3.8	-3.4	1.4	2.6
D9000X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	12.0	2.4	2.7	5.6	2.9	3.2	0	0	0	0	0	0	-12.0	-2.4	-2.7	-5.6	-2.9	-3.2
D9200X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0
D9000 Complex	Change	Revision	1/1/2017 URCR Update	0	0	0	0	0	0	14.6	2.2	7.8	4.8	2.7	6.2	14.6	2.2	7.8	4.8	2.7	6.2
E1000X	Change	Revision	1/1/2017 URCR Update	29.0	7.8	8.8	20.3	5.5	6.2	27.9	12.3	37.3	8.9	8.0	23.4	-1.1	4.5	28.5	-11.4	2.5	17.2
D1000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0

Restricted

D4000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D5000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D6000M	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D6000X	No Change	NA	NA	89.9	15.2	16.5	53.5	9.4	10.1	89.9	15.2	16.5	53.5	9.4	10.1	0.0	0.0	0.0	0.0	0.0	0.0
D7000X	No Change	NA	NA	1.7	0.8	1.0	-	-	-	1.7	0.8	1.0	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
D7100X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
E3000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0
G1000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0

Table 2D: Associated Gas in-Place & Ultimate Recovery Changes - NNS P1-P2-P3 Method (Bscf)

				1/1/2016 ARPR						1/1/2017 ARPR						Change During 2016					
				Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery		
Reservoir	Reservoir Changes	Reasons for Change	Reference Document	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
D8000X	Change	Revision	1/1/2017 URCR Update	21.1	0.0	0.0	7.7	0.1	0.0	8.2	7.8	12.2	4.3	4.7	8.4	-12.9	7.8	12.2	-3.4	4.6	8.4
D9000X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	28.5	6.0	6.6	17.3	2.9	3.2	0	0	0	0	0	0	-28.5	-6.0	-6.6	-17.3	-2.9	-3.2
D9200X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	9.8	1.7	1.8	4.5	0.8	0.8	0	0	0	0	0	0	-9.8	-1.7	-1.8	-4.5	-0.8	-0.8
D9000 Complex	Change	Revision	1/1/2017 URCR Update	0	0	0	0	0	0	29.3	20.4	31.4	13.1	15.2	24.5	29.3	20.4	31.4	13.1	15.2	24.5
E1000X	Change	Revision	1/1/2017 URCR Update	44.4	10.0	8.8	43.6	21.7	13.2	47.1	28.9	56.5	20.1	19.7	37.9	2.7	18.9	47.7	-23.5	-2.0	24.7
D1000X	No Change	NA	NA	2.6	0.1	0.2	3.3	0.0	0.0	2.6	0.1	0.2	3.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Restricted

D4000X	No Change	NA	NA	0.3	0.2	0.3	0.1	0.0	0.0	0.3	0.2	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D5000X	No Change	NA	NA	0.3	0.2	0.3	0.1	0.0	0.0	0.3	0.2	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D6000M	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D6000X	No Change	NA	NA	94.1	15.9	17.2	53.5	9.4	10.1	94.1	15.9	17.2	53.5	9.4	10.1	0.0	0.0	0.0	0.0	0.0	0.0
D7000X	No Change	NA	NA	2.0	0.8	1.3	0.0	0.0	0.0	2.0	0.8	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D7100X	No Change	NA	NA	0.8	0.0	0.1	0.0	0.0	0.0	0.8	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
E3000X	No Change	NA	NA	0.3	0.1	0.2	0.0	0.0	0.0	0.3	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
G1000X	No Change	NA	NA	42.1	0.0	0.0	24.6	0.0	0.0	42.1	0.0	0.0	24.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 2E: Associated Condensate in-Place & Ultimate Recovery Changes - NNS P1-P2-P3 Method (MMstb)

Reservoir	Reservoir Changes	Reasons for Change	Reference Document	1/1/2016 ARPR						1/1/2017 ARPR						Change During 2016					
				Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery		
				P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
D8000X	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0030	0.0070	0.0100	0.0012	0.0037	0.0064	0.0	0.0	0.0	0.0	0.0	0.0
D9000X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	-	-	0.1	-	-	-	0	0	0	0	0	0	-	-	-0.1	-	-	-
D9200X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0
D9000 Complex	Change	Revision	1/1/2017 URCR Update	0	0	0	0	0	0	0.0600	0.0100	0.0300	0.0200	0.0100	0.0200	0.0600	0.0100	0.0300	0.0200	0.0100	0.0200
E1000X	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.2700	0.1100	0.3600	0.0900	0.0700	0.2200	0.27	0.11	0.36	0.09	0.07	0.22
D1000X	No Change	NA	NA													0	0	0	0	0	0

Restricted

				-	-	-	0.4	0.4	0.4	-	-	-	0.4	0.4	0.4						
D4000X	No Change	NA	NA	0.1	0.1	0.1	0.1	-	-	0.1	0.1	0.1	0.1	-	-	0	0	0	0	0	0
D5000X	No Change	NA	NA	0.3	0.3	0.3	0.1	0.2	0.1	0.3	0.3	0.3	0.1	0.2	0.1	0	0	0	0	0	0
D6000M	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0
D6000X	No Change	NA	NA	-	0.1	-	0.1	-	-	-	0.1	-	0.1	-	-	0	0	0	0	0	0
D7000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0
D7100X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0
E3000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0
G1000X	No Change	NA	NA	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Table 2F: Non-Associated Gas in-Place & Ultimate Recovery Changes - NNS P1-P2-P3 Method (Bscf)

Reservoir	Reservoir Changes	Reasons for Change	Reference Document	1/1/2016 ARPR						1/1/2017 ARPR						Change During 2016					
				Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery		
				P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
D8000X	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D9000X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D9200X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D9000 Complex	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
E1000X	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D1000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Restricted

D4000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D5000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D6000M	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D6000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D7000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D7100X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
E3000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
G1000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 2G: Non-Associated Condensate in-Place & Ultimate Recovery Changes - NNS P1-P2-P3 Method (MMstb)

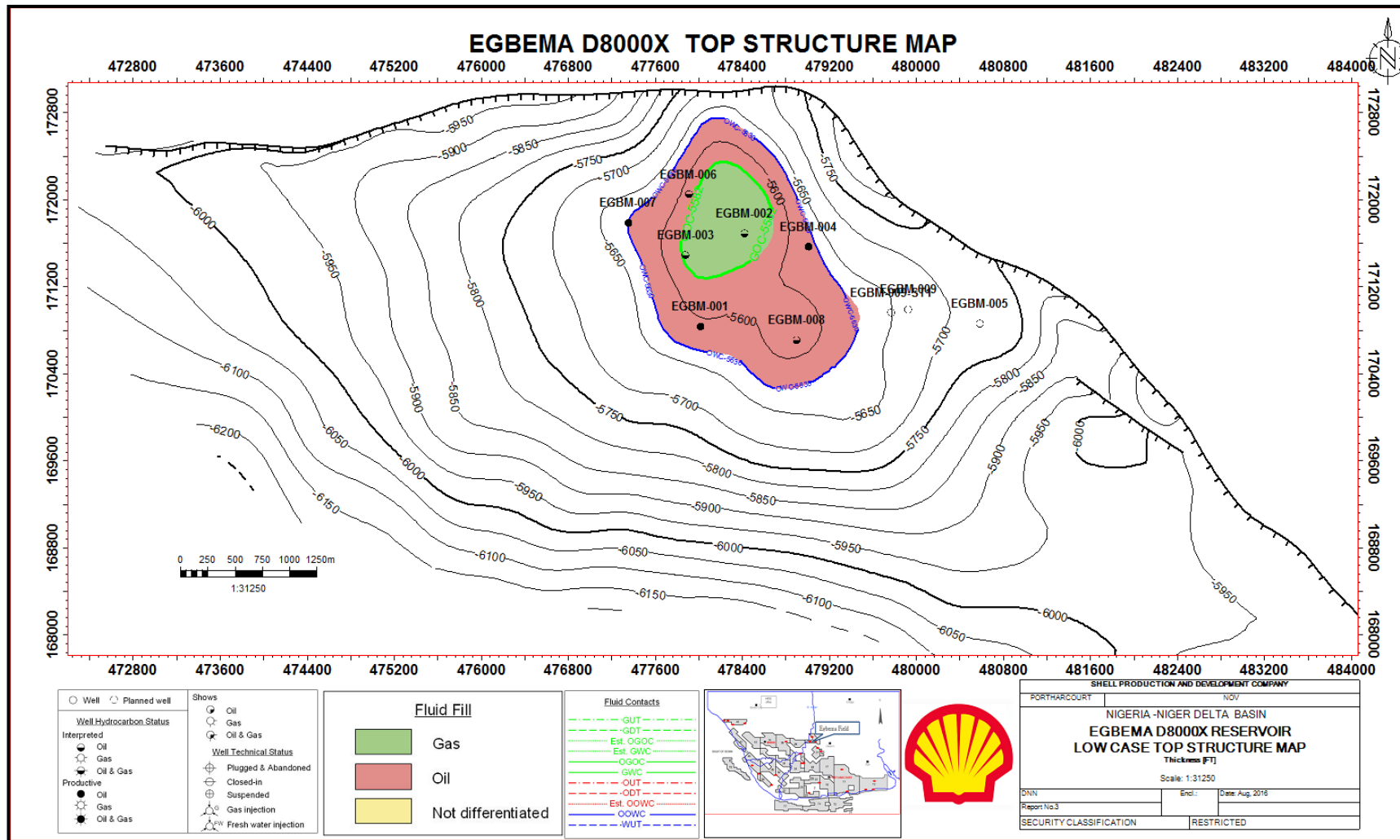
				1/1/2016 ARPR						1/1/2017 ARPR						Change During 2016					
				Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery			Hydrocarbon-in-place			Ultimate Recovery		
Reservoir	Reservoir Changes	Reasons for Change	Reference Document	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
D8000X	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D9000X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D9200X	Delete	Merged D9,D91 & D92 as D9 complex	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D9000 Complex	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
E1000X	Change	Revision	1/1/2017 URCR Update	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D1000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D4000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Restricted

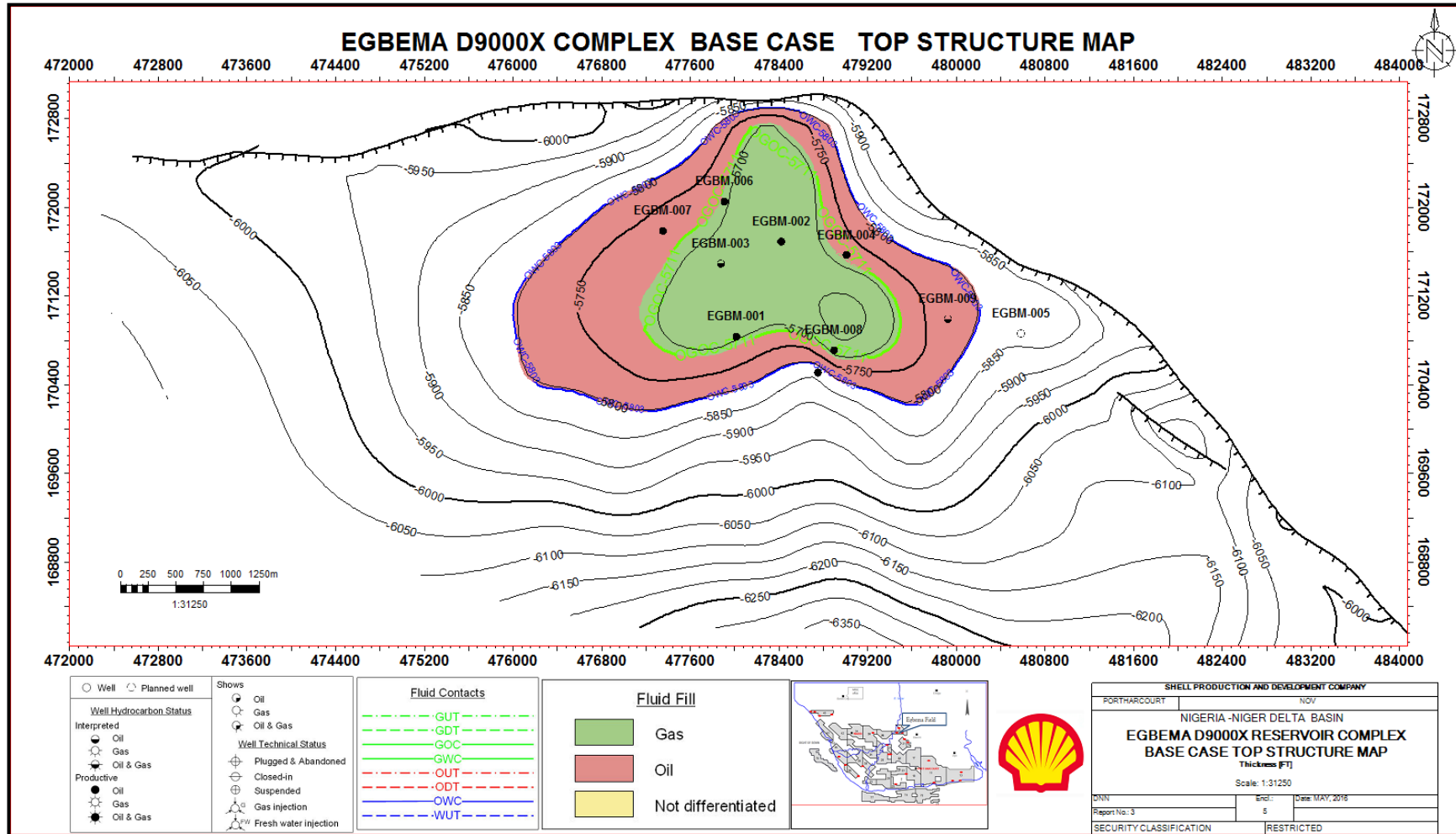
D5000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D6000M	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D6000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D7000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
D7100X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
E3000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
G1000X	No Change	NA	NA	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

A1.6. Maps

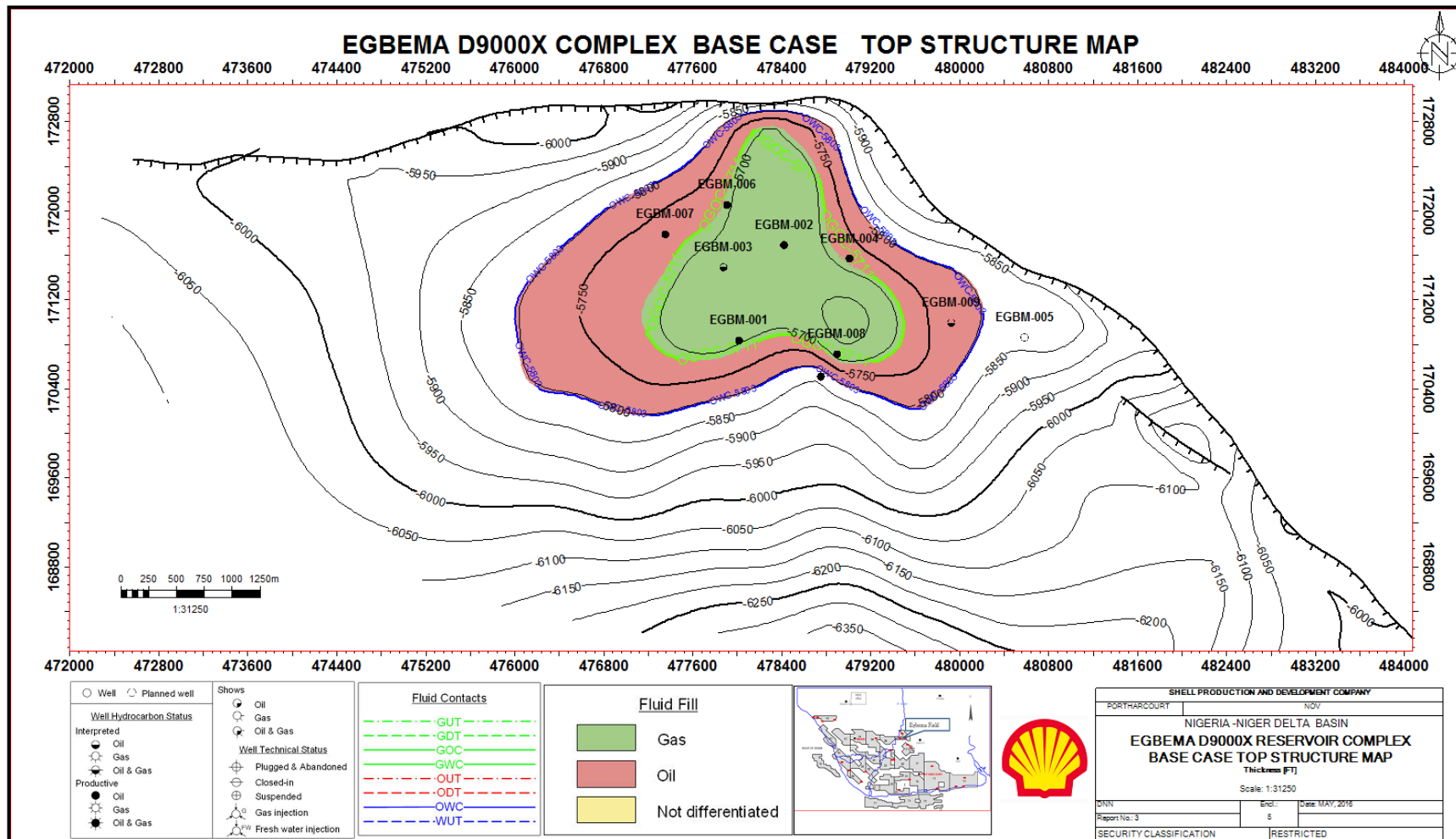
A1.6.a Egbema D8000X Top Structure Map

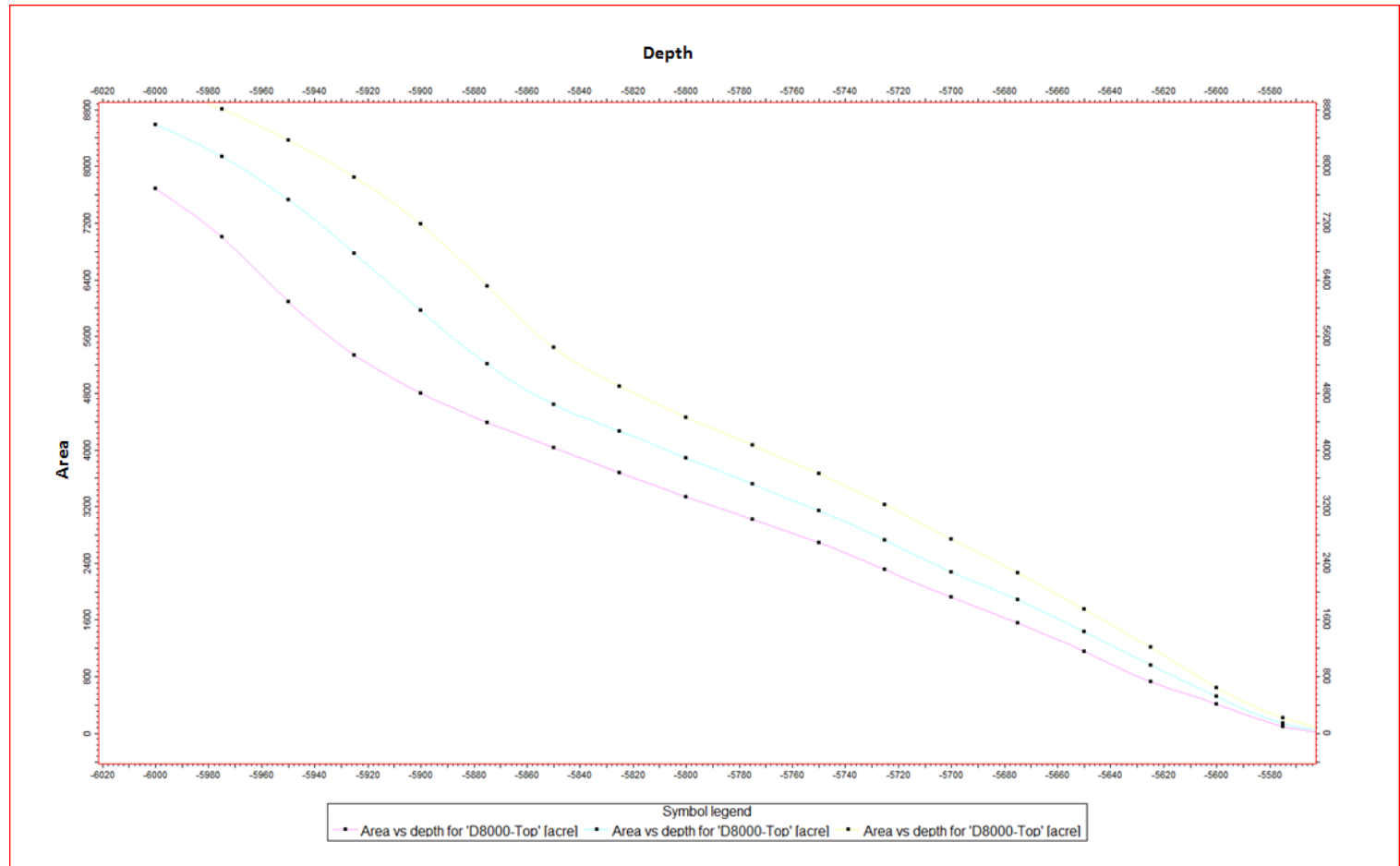


A1.6.b Egbema D9000X Top Structure Map

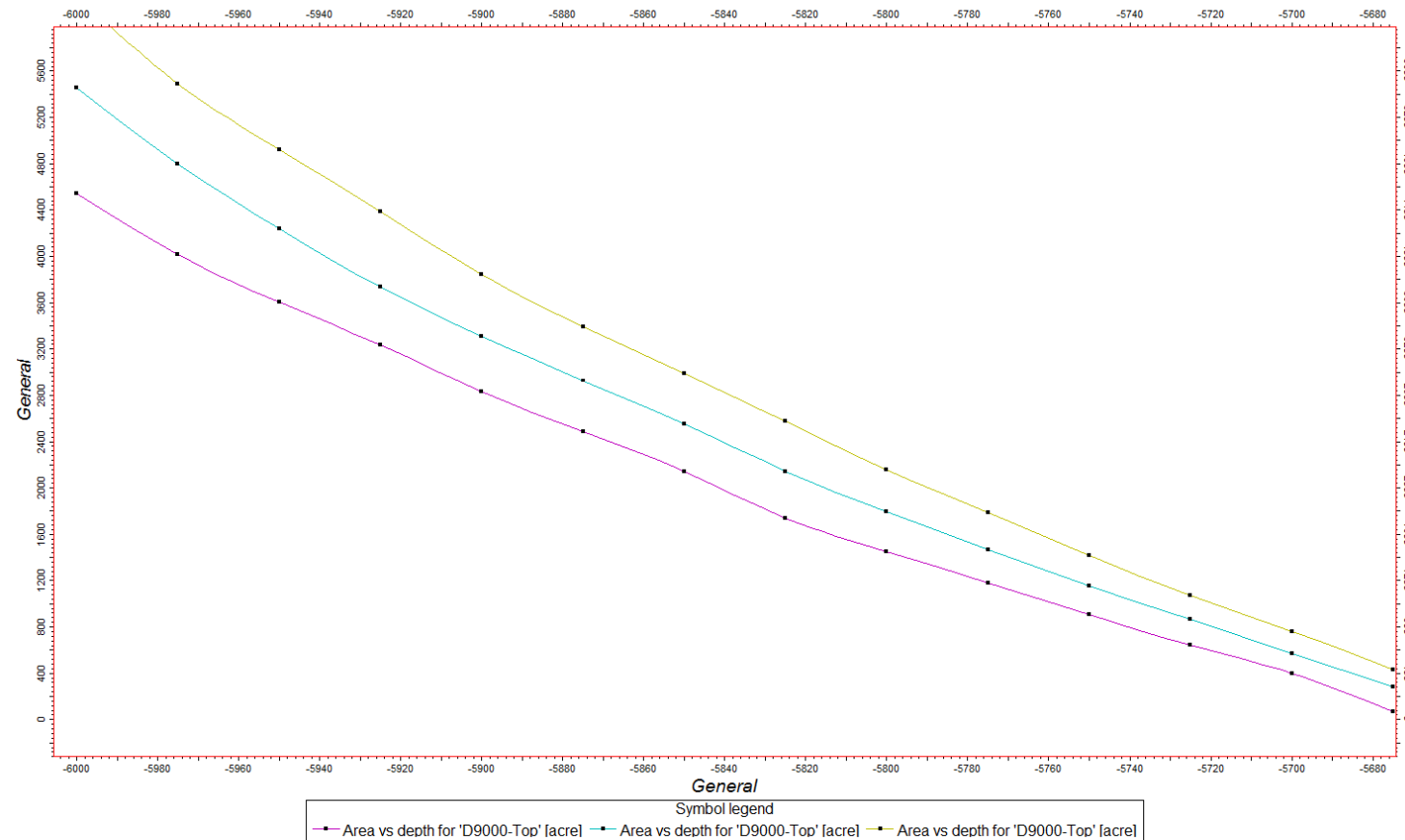


A1.6.b Egbema E1000X Top Structure Map



A1.7. Area-Depth Plots**A1.8. A1.7.a Egbema D8000X Reservoir Area-Depth Plots**

A1.7.b Egbema D9000X Reservoir Area-Depth Plots



A1.7.c Egbema E1000X Reservoir Area-Depth Plots

